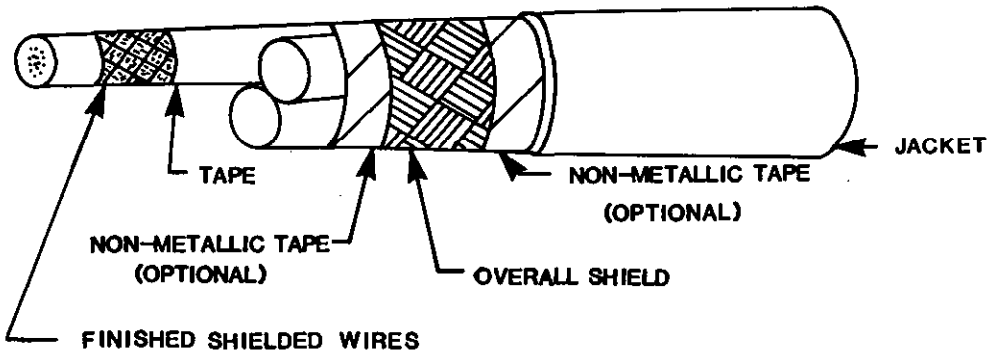
MILITARY SPECIFICATION SHEET

# CABLE, ELECTRICAL, 600 VOLTS, LIGHTWEIGHT, MULTI-CONDUCTOR, AWG 22, TYPE 1XSOW

This military specification has been prepared by the Naval Sea Systems Command based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-C-24640. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24640.




SH 12761

FINISHED SHIELDED WIRE:

First - Copper conductor, size AWG 22, 7 stranding. Nominal diameter of stranded conductor 0.030.
Second - Insulation material requirements in accordance with MIL-W-81044/12 except the conductor and identification codes and methods shall be in accordance with MIL-C-24640. Finished wire diameter 0.044 inch minimum and 0.050 inch maximum.
Third - Shield shall be tin-coated copper braid with a coverage not less than 85 percent determined in accordance with MIL-C-27500.

FIGURE 1. Construction (watertight).

(A) denotes changes.

FSC 6145

DISTRIBUTION STATEMENT A Approved for public release; distribution unlimited



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(A) TABLE I. Construction details.

| Type number | Number of conductors | Diameter over shield (inch) (nom) | Jacket thickness (inch) (minimum) | Cable diameter (inch) (min) | Cable diameter (inch) (max) | Cable weight (pounds per 1000 feet) (max) |
|---|---|---|---|---|---|---|
| 1XSOW-2 | 2 | 0.173 | 0.034 | 0.292 | 0.314 | 83.0 |
| 1XSOW-14 | 14 | .362 | .041 | .470 | .506 | 233.0 |
| 1XSOW-20 | 20 | .429 | .044 | .542 | .584 | 307.0 |
| 1XSOW-30 | 30 | .520 | .044 | .614 | .662 | 429.0 |

REQUIREMENTS:

Qualification required.
Construction details: See table I.

Overall shield shall be tin-coated copper braid with a coverage not less than 85 percent determined in accordance with MIL-C-27500.

EXAMINATION AND TESTS:

| | Requirements |
|---|---|
| Basic electrical: | |
| Conductor resistance - ohms/1000 feet, at 25°C, maximum | 17.03 |
| Voltage withstand - volts, root mean square, minimum | |
| Conductor to conductor | 2500 |
| Conductor to shield | 1000 |
| Shield to shield | 500 |
| Insulation resistance - megohms/1000 feet, minimum | 5000 |
| Conductor and shield continuity | No failure |
| Jacket flaws | No failure |
| Group A: | |
| Visual and dimensional | No failure |
| Capacitance: | |
| Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft), maximum | 40 |
| Capacitance unbalance - percent, maximum | 15 |
| Characteristic impedance - at 1 MHz, ohms | 60 ± 5 |
| Watertightness (see table XI of MIL-C-24640 for limits of water leakage) | No failure |
| Weight | No failure |



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EXAMINATION AND TESTS: (Continued)

| | Requirements |
|---|---|
| Group B: | |
| Cold bending | No failure |
| Drip | Zero |
| Tear - pounds per inch thickness, minimum | 35 |
| Physicals (unaged) | |
| Jacket | |
| Tensile strength - $lb/in^2$, minimum | 1300 |
| Elongation - percent, minimum | 160 |
| Attenuation - at 3 MHz, dB/100 feet, maximum | 3 |
| Gas flame (1 hour) | Not applicable |
| Group C: | |
| Physicals (aged) | |
| Jacket | |
| Air oven | |
| Tensile strength - percent of unaged, minimum | 60 |
| Elongation - percent of unaged, minimum | 60 |
| Hot oil immersion | |
| Tensile strength - $lb/in^2$, minimum | 50 |
| Elongation - percent, minimum | 50 |
| Permanence of printing (conductor) - cycles, minimum | 50 |
| Permanence of printing (jacket) - cycles, minimum | 250 |
| Armor - conformance to material, construction and coverage | Not applicable |
| Shield - conformance to material, construction and coverage | No failure |
| Heat distortion - percent unaged, maximum | 30 |
| Cable filler removability | No failure |
| Shrinkage | No failure |
| Mutual inductance - dB/1000 feet, maximum | Not applicable |
| Surface transfer impedance | |
| Milliohms per meter, maximum | 700 |
| EMP response, dB, minimum | 60 |
| Pulse response time - ms/500 feet | |
| Rise time, maximum | Not applicable |
| Decay time, maximum | Not applicable |



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QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

| | Requirements |
|---|---|
| Aging and compatibility (cable) | No failure |
| Acid gas equivalent - percent, maximum | |
| Jacket | 2 |
| Fillers | 2 |
| Insulation | 18 |
| Halogen content - percent, maximum | |
| Jacket | 0.2 |
| Fillers | 0.2 |
| Flame propagation | No failure |
| Immersion | |
| Tensile strength - percent of unaged, minimum | 50 |
| Elongation - percent of unaged, minimum | 50 |
| Smoke index, maximum | |
| Jacket | 25 |
| Fillers | -- |
| Insulation | 10 |
| Toxicity index, maximum | |
| Jacket | 5 |
| Fillers | 5 |
| Insulation | 1.5 |
| Durometer - (type A) hardness, minimum | 80 |
| Weathering | No failure |
| Fungus resistance | No failure |
| Abrasion resistance | No failure |
| Gas flame (3 hour) | Not applicable |
| Finished wire | Not applicable |

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